



Deep learning-based extension of wind pressure time series

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ABSTRACT

The spatio-temporal variation of the wind pressure field is crucial for understanding structural loads and their effect on design. However, obtaining long-duration wind pressure time series around bluff bodies through wind tunnel tests or stochastic and computational simulations is both costly and time-consuming. To address this challenge, this study develops a deep learning (DL) model called WPTSE-Net for extending non-Gaussian wind pressure time series, thereby eliminating the need for the characterization of their nonlinear features and providing an end-to-end flexible framework for extending pressure coefficient time series. The key innovation of WPTSE-Net lies in the reconstruction of the encoder, utilizing prior knowledge to eliminate complex steps in searching for the latent space. This improvement not only enhances computational efficiency and model performance but also substantially reduces the amount of training data that is required for the DL generative model. Comparative results indicate that the proposed WPTSE-Net model outperforms traditional methods in terms of statistical characteristics, i.e., spectra, and peak value distributions. Thus, WPTSE-Net is highly suitable for practical engineering applications as it provides an efficient means of generating long-time series of wind pressure on bluff bodies in wind resistance design.

1. Introduction

Wind loading is a stochastic process, and it cannot be adequately characterized by simple statistical features alone. To obtain a more comprehensive understanding of the load requires collecting data over an extended period. This is especially crucial for specific applications, including the assessment of extreme values, structural health monitoring, and fatigue analysis. For instance, Holmes and Cochran (2003) estimated upper limit extreme values based on thousands of wind pressure time series obtained through wind tunnel experiments. Repetto and Torrielli (2017) conducted a 50-year fatigue analysis for slender structures under wind loading. Additionally, determining annual extreme wind speeds requires extensive observation spanning several decades, involving the compilation of a database capturing the maximum wind speeds each year for subsequent analysis and assessment. However, due to limitations in testing conditions and time costs, obtaining extensive wind pressure data over long durations (large samples) through on-site measurement, wind tunnel experiments, stochastic simulation, or CFD simulations is challenging. An effective strategy for acquiring long-term wind pressure processes involves using short yet

reliable data to simulate/extend/predict time series data. The primary advantage of this alternative approach lies in the significant reduction of resources and time needed for long-term measurement or simulation. It offers a convenient means to understand the entire non-Gaussian characteristics and describe the time-domain information of wind pressure, making it highly practical for engineering applications.

Numerous researchers have explored various methods for simulating and extending wind pressure time series (Kareem, 2020; Salehinejad and Flay, 2021), including static transformation methods, memory-based transformation methods, wavelet transforms, probability density function (PDF) modeling, and among others (Gurley et al., 1996; Gurley and Kareem, 1997, 1998). Static transformation methods, such as correlation distortion simulation, modified direct simulation, and spectral correction simulation, are commonly used due to their clear conceptual framework and efficient computation. However, these methods are limited by their inability to preserve phase interactions among related frequency components, which can lead to inaccuracies in simulations involving complex wind pressure patterns. Memory-based methods, like the Volterra series model and NARMAX (non-linear autoregressive

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moving average models with exogenous input), are typically data-driven. These methods often suffer from error accumulation, particularly when applied to non-stationary data over extended periods. This limitation makes them less effective for capturing dynamic behaviors in ultra-long simulations. Wavelet transform methods, while powerful, are highly dependent on the choice of wavelet function and can be computationally intensive. PDF methods also have their drawbacks, particularly in handling trailing data. In practical engineering applications, correlation distortion simulation remains widely used for its straightforward concept and computational efficiency. Recent studies have attempted to enhance this method to improve its applicability and accuracy (Kumar and Stathopoulos, 1997, 1999; Gioffre et al., 2000; Gioffre et al., 2001; Yang and Gurley, 2015; Yang and Tian, 2015; Liu et al., 2017, 2020; Wu et al., 2020, 2022). Nevertheless, the simulation accuracy of correlation distortion static transformation may be constrained by the autocorrelation function and distorted higher-order information, and it exhibits suboptimal performance in strongly non-Gaussian processes, such as wind pressure at the corners of buildings.

In recent years, machine learning (ML) techniques have been introduced to the field of wind engineering (Hu and Kwok, 2020; Arul et al., 2020; Lin et al., 2022; Arul et al., 2022; Deng et al., 2024; Liu et al., 2023). Deep learning (DL), at the forefront of ML advancements, stands as a powerful tool for modeling across various research domains, capitalizing on its efficiency, adaptability, and accuracy. Substantial research has been conducted to date on methods for predicting wind speed/pressure sequence data, including Deep Neural Networks (DNN), Backpropagation Neural Networks (BPNN), Wavelet Neural Networks (WNN), Long Short-Term Memory (LSTM), Convolutional Long Short-Term Memory (ConvLSTM), Temporal Convolutional Network (TCN), Graph Neural Network (GNN), Transformer, among others (Huang et al., 2017; Fernández-Cabán et al., 2018; Snaiki and Wu, 2019; Luo and Kareem, 2020; Li et al., 2022; Chen et al., 2022; Tong et al., 2022; Jiang et al., 2023b; Liu et al., 2023; Jiang et al., 2023a; Gao et al., 2023, 2024a; Zhang et al., 2024). However, these prediction methods do have certain limitations, particularly an accumulation of errors as the prediction range extends, especially for long sequences. Additionally, gradient vanishing and exploding issues may arise for ultralong time sequences. Furthermore, these prediction algorithms consistently encounter challenges in non-Gaussian and non-stationary data.

In the domain of DL, there is a category of methods specifically designed to generate/extend new data from existing samples, commonly referred to as generative models. This category encompasses Variational Auto-Encoder (Kingma and Welling, 2013), Normalizing Flows (Rezende and Mohamed, 2015), Generative Adversarial Networks (Goodfellow et al., 2014), and Diffusion models (Ho et al., 2020), etc. Currently, these generative models are primarily applied to wind pressure super-resolution (Hu et al., 2020; Gao et al., 2024b,a), with limited exploration in non-Gaussian fluctuating wind pressure simulations. The prevailing situation is primarily attributed to challenges associated with handling long-time series data, leading to unstable performance and issues such as gradient vanishing and mode collapse. Consequently, these challenges result in the loss of temporal characteristics within the data. Furthermore, generative models often exhibit a bias toward learning smooth distributions, posing challenges in accurately capturing abrupt changes and non-stationary relationships inherent in real data. Of greater significance, the intricate structures and substantial parameter counts of generative models render their performance heavily dependent on the quantity of available data. In most cases, these data are relatively scarce or not easily accessible.

In this regard, the present study proposes a specialized generative model, the Wind Pressure Time Series Extend Network (WPTSE-Net), designed for extending time series data. WPTSE-Net stands out for its unique integration of prior knowledge, significantly reducing the

required data while maintaining high performance. It not only effectively extends short-sequence wind pressure data for arbitrary bluff body shapes but also has universality across various data sources, such as measurements or experiments. Furthermore, the model's simplicity, flexibility, and efficiency make it particularly well-suited for low-dimensional data, enabling seamless end-to-end data generation and reducing time and resource consumption. Therefore, it provides a novel perspective and serves as a valuable reference for assessing wind pressure on building surfaces in engineering practice.

The structure of this paper is organized as follows: Section 2 introduces the data source, and methods employed, and outlines the extended procedure and evaluation methods. Section 3 presents the extension results, providing a detailed comparison of various perspectives. The conclusions are drawn in Section 4.

2. Data and methodology description

2.1. Wind pressure data source

The data used in this study primarily originated from the WD-1 wind tunnel laboratory at Wuhan University, including wind pressure coefficients (C_p) measured from two rounded-corner tall building models. Both test models were 1.2 meters in height and 0.3 meters in side length, with a length scale of 1:200 relative to the actual building. The first building (Model I) possessed rounded corners with a radius of 15 millimeters, whereas the second building (Model II) featured corners with a radius extended to 45 millimeters. Each Model is equipped with three pressure measurement points, positioned at a height of 1.07 m. We designated the three points of Model I as points 1 to 3, and the points of Model II as points 4 to 6, following the labeling illustrated in Fig. 1. In this experiment, a reference wind speed of 10 m/s in a category C wind field (Chinese Code GB 50009-2012) was chosen for the calibration of wind profiles of the oncoming flow, and the wind attack angle was set as 0° .

After data collecting and processing, the total available duration of the wind pressure data provided by Model I was 200 min, while Model II provided a total of 500 min, as shown in Fig. 2. Drawing on the World Meteorological Organization's (WMO) recommendation to use a ten-minute average wind speed standard (Harper et al., 2010), and considering engineering practices that account for the sustained wind loads on structures, we have segmented the wind pressure data in this study into ten-minute intervals. Both model datasets are divided accordingly, resulting in 20 segments for the Model I dataset and 50 segments for the Model II dataset. One of the ten-minute segments contains 4096 values (sampling frequency is approximately 7 Hz). In this study, only one ten-minute segment (depicted by the gray line in Fig. 2) was used for training the deep learning model or traditional methods. The remaining data segments were designated as ground truth (represented by the red line in Fig. 2) and were used for comparison and analysis against the extended results during the testing phase.

2.2. Algorithm

2.2.1. Wind pressure time series extend network (WPTSE-Net)

The Wind Pressure Time Series Extend Network (WPTSE-Net) is a unique generative model specifically developed for extending or simulating C_p time-series data. Typically, generative models comprise two components: Encoder and Decoder. The Encoder transforms high-dimensional data into a compact low-dimensional vector, often referred to as latent space or latent vector. Next, the Decoder converts the latent vector into new synthesized data. The latent space can be viewed as a low-dimensional abstract representation space, where each point corresponds to an abstract feature or concept learned by the neural network. The expressive capacity of this latent space is crucial for the performance of generative models and the quality of generated samples. In the structure of WPTSE-Net, we control the Encoder section,

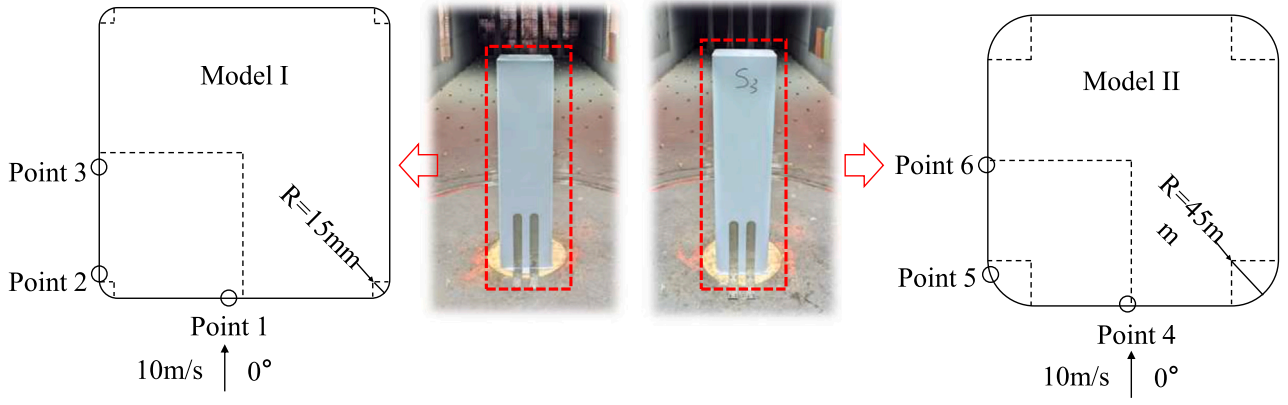


Fig. 1. Two wind tunnel models and schematic depicting the locations of pressure measurement points.

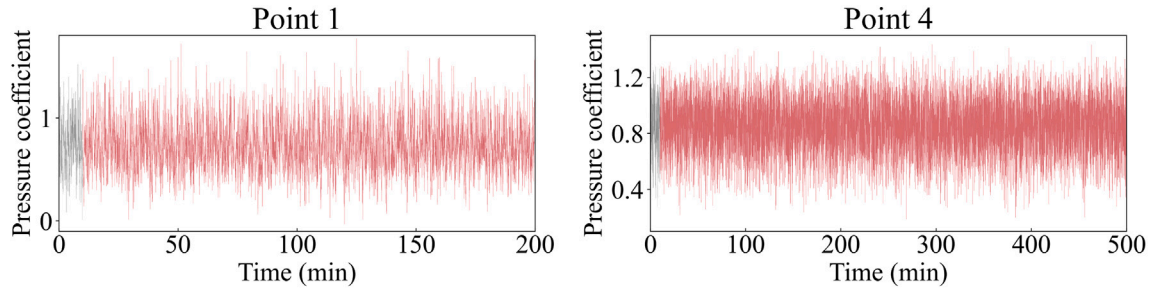


Fig. 2. Full-length wind tunnel experimental data of point 1 and point 4.

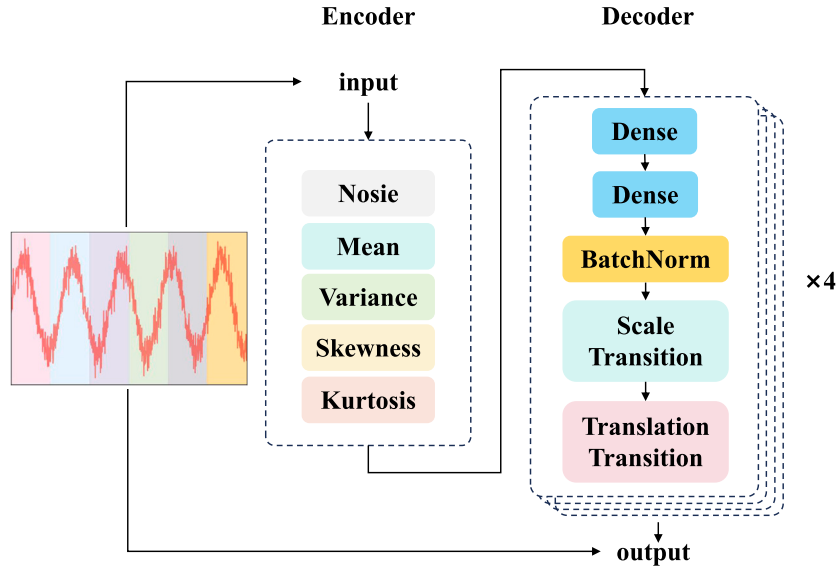


Fig. 3. Schematic diagram illustrating the structure of the WPTSE-Net.

eliminating the need for the generative model to search for latent vectors autonomously. Instead, we manually design/set latent vectors with physical or mathematical significance. This approach not only introduces prior knowledge but also avoids the possibility of obtaining ineffective latent spaces. In contrast to advanced and complex generative models, WPTSE-Net not only avoids substantial data-driven requirements but also features a lighter and more flexible framework, particularly well-suited for simple and low-dimensional data.

The structure schematic diagram of WPTSE-Net is illustrated in Fig. 3. WPTSE-Net comprises two computational modules: a unique Encoder and a Decoder. In Encoder, we define the latent space as

statistical features of output sequences and introduce noise encoding. Specifically, we select the mean, variance, skewness, and kurtosis as the representatives, combined with noise encoding to form a latent vector of l rows and five columns (where l represents the number of slices in the datasets, refer to Section 2.3). The primary role of noise encoding is to import disturbance to prevent model overfitting and to improve sample diversity. Once the Encoder is complete, the input proceeds to the Decoder for computation. The Decoder module consists of four iterations, as indicated by the '×4' designation in Fig. 3. Each iteration block includes two dense layers, one regularization operation, one translation layer (multiplicative), and one scale transformation

layer (additive). Two dense layers are responsible for feature expression and mapping, containing 128 neurons in conjunction with RELU activation. The regularization layers facilitate enhanced model generalization capabilities, thus optimizing its performance in practical applications. The iteration block is further refined through the application of scaling and translation operations (Dinh et al., 2016), which serve to enhance sample diversity and reinforce the stability and robustness of model training. The aforementioned layers execute operations on the data in a serial manner. The final pass data is directed to a linear full-connected layer, generating new slices of wind pressure sequences. In this study, the optimization function uses Adam with a batch size of 64, trained 1000 times.

Finally, the generated sequences and label sequences undergo error calculation using a specified method, i.e., the loss function. By feeding back error information, the network parameters are iteratively adjusted to achieve optimal values (i.e., backpropagation). In this study, we opt for Huber loss as the loss function, which inherently balances Mean Squared Error (MSE) and Mean Absolute Error (MAE). Its formula is as follows:

$$L_{\delta}(y, f(\hat{x})) = \begin{cases} \frac{1}{2}(y - f(\hat{x}))^2, & |y - f(\hat{x})| \leq \delta \\ \delta|y - f(\hat{x})| - \frac{1}{2}\delta^2, & |y - f(\hat{x})| > \delta \end{cases} \quad (1)$$

in which y is the true label, $\hat{x}$ is the model's input, $f(\hat{x})$ is the model's output, δ is a predefined threshold, and here δ is set to 1. L_{δ} represents the Huber loss with threshold δ .

The data addressed in this study are low-dimensional time-series data and are relatively scarce. In fact, advanced generative models like Variational Auto-Encoder and Generative Adversarial Networks could be applied to wind pressure time series data. However, our preliminary experiments indicate that results from these complex models are notably inferior, even incapable of capturing basic mean and variance aspects.

Given that highly complex network structures may lead to unnecessary model complexity and prolonged calculation times, we adopt conventional computational modules to better adapt to data features while maintaining a concise and efficient model structure. The results of our experiments demonstrate that WPTSE-Net requires approximately 5.5 s to generate the 500-min data of Model II using the NVIDIA GeForce RTX 3090 GPU, a significantly faster process than that typically observed in the GANs model (such as DCGAN) that often requires several tens of seconds or even longer. It is worth noting that WPTSE-Net has the potential to integrate more complex and advanced computational modules, such as LSTM, attention mechanisms, and residual networks, particularly when dealing with high-dimensional data like images or videos.

2.2.2. Two classic approaches

(i) Correlation Distortion Static Transform (CDST)

Among classic methods, the Correlation Distortion Static Transform (CDST) method is widely employed due to its conceptual straightforwardness and efficiency, particularly in simulating non-Gaussian processes (Gurley et al., 1997). This method begins with the autocorrelation function of the non-Gaussian process $R_{xx}(\tau)$ instead of a sample of a non-Gaussian time history x_t (raw data). It not only improves the matching between the simulated non-Gaussian signal and the non-Gaussian spectrum of the samples but also eliminates the need for an iterative process.

First, the autocorrelation function of the Gaussian process $R_{xx}(\tau)$ is calculated according to Eq. (2).

$$R_{xx}(\tau) = \frac{\sum_{t=1}^{n-\tau} (x_t - \bar{x})(x_{t+\tau} - \bar{x})}{\sum_{t=1}^n (x_t - \bar{x})^2} \quad (2)$$

Next, start from the autocorrelation function of the non-Gaussian process $R_{xx}(\tau)$ and transform it to obtain the autocorrelation function of

the underlying Gaussian process $R_{uu}(\tau)$. Specifically as shown in Eq. (3) to (6):

$$R_{xx}(\tau) = \alpha^2 [R_{uu}(\tau) + 2\hat{h}_3^2 R_{uu}^2(\tau) + 6\hat{h}_4^2 R_{uu}^3(\tau)] \quad (3)$$

$$\hat{h}_3 = \frac{\gamma_3}{4 + 2\sqrt{1 + 1.5\gamma_4}} \quad (4)$$

$$\hat{h}_4 = \frac{\sqrt{1 + 1.5\gamma_4} - 1}{18} \quad (5)$$

$$\alpha = \frac{1}{\sqrt{1 + 2\hat{h}_3 + 6\hat{h}_4^2}} \quad (6)$$

where γ_3 and γ_4 are the skewness and kurtosis of the fluctuating process, x_t stand for Gaussian process/values, $\bar{x}$ denote the mean of x_t .

Once the underlying Gaussian process $R_{uu}(\tau)$ was completed, a Fourier transform was performed on the autocorrelation function to obtain the spectrum $G_{uu}(\omega)$. Finally, utilize the formula to transform the Gaussian process u into a non-Gaussian process. Thus, this method obtains the simulation of the non-Gaussian process with zero mean and unit variance. Performing a simple linear transformation on this simulation yields the final result. As shown in Eq. (7) to (8):

$$x_{nor} = a \left[u + \hat{h}_3(u^2 - 1) + \hat{h}_4(u^3 - 3u) \right] \quad (7)$$

$$Y_e = \sigma(x_{nor} + \mu) \quad (8)$$

where μ and σ are the mean and standard deviation of the fluctuating process x_t .

Numerous subsequent studies have generated diverse enhanced versions of CDST due to its remarkable practical application. Notably, certain of these improved variants have exhibited promising performance in wind pressure assessment. Estimation of the surface pressure extremes using hybrid data and simulation-based approach presented in Ma et al. (2016) utilizes the concept of extending time series based on a combination of an AR and inverse Johnson transformation. This approach demonstrated its suitability for both softening and hardening type stationary non-Gaussian processes for any combination of skewness and kurtosis.

(ii) Gaussian Kernel Density Estimation (GKDE)

As a non-parametric method for estimating probability density, Kernel Density Estimation is widely employed to infer probability density from data. Within the methodology of kernel density estimation, the Gaussian kernel density estimation (GKDE) stands out as a common kernel function (Parzen, 1962; Davis et al., 2011). It possesses the advantage of adapting to various shapes of data distributions and provides a smooth density estimation. Particularly in situations lacking prior distribution information, it can effectively estimate the probability density function of the data. Therefore, Gaussian kernel density estimation is chosen as one of the conventional simulation methods in this study. By fitting the probability density of C_p and conducting sampling, it extends arbitrary amounts of time-series data.

The fundamental principle of GKDE involves using the Gaussian distribution as the kernel function. This entails calculating the weight for each data point and aggregating these Gaussian functions to obtain the probability density function estimation for the entire dataset. The mathematical expression for the Gaussian kernel function is presented in Eq. (9). For one-dimensional data, the estimation of the probability density function can be expressed by Eq. (10).

$$K(u) = \frac{1}{\sqrt{2\pi}} e^{-\frac{u^2}{2}} \quad (9)$$

$$\hat{f}(x_{nor}) = \frac{1}{n \cdot h} \sum_{i=1}^n K\left(\frac{x_{nor} - x_i}{h}\right) \quad (10)$$

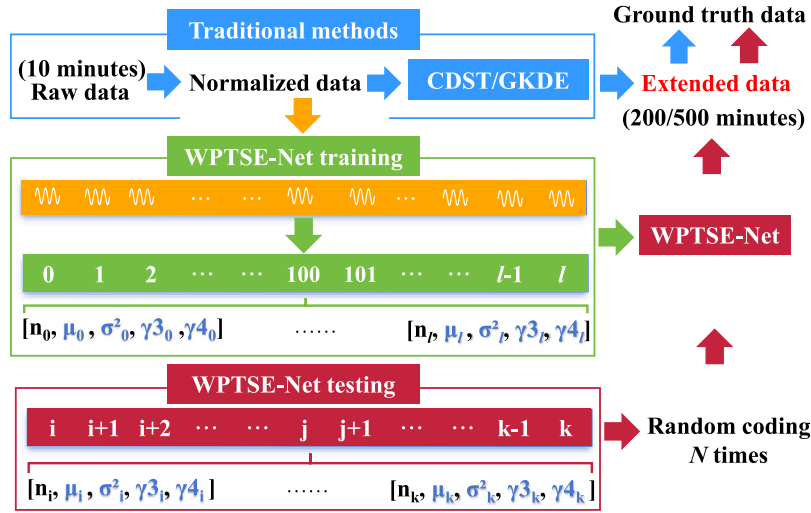


Fig. 4. Flowchart illustrating the time series extension procedure.

in the equation, $\hat{f}(x_{nor})$ represents the probability density estimation at x_{nor} , n is the data length, h denotes the bandwidth, and x_i represents each data point in the datasets. The bandwidth is a parameter controlling the width of the kernel and its choice significantly impacts the estimation results. A cross-validation method was employed to determine the most appropriate bandwidth for the data, resulting in a value of 0.1.

2.3. Time series extension procedure

Fig. 4 illustrates our specific step-by-step process of the data extension procedure. First of all, the raw data of the first ten-minute are processed by applying the normalization transformation. After the data standardization step, the normalized data is treated as input. Methods of CDST and GKDE are then employed to generate extended data. However, for WPTSE-Net, additional segmentation and processing are required.

During the WPTSE-Net procedure, the training segment (a ten-minute normalized data) is partitioned into l slices, each containing $4096/l$ values. This implies that the training set comprises a total of l inputs and outputs (labels), each output is a sequence of $4096/l$ values. The input for the WPTSE-Net is calculated based on the corresponding output label and comprises four features: [mean, variance, skewness, kurtosis]. Additionally, we define that the 'noise code' follows a normal distribution and is multiplied by a scaling factor of 0.1. In the testing phase, new noise encoding is generated for the input, and all inputs are randomly shuffled. Subsequently, the corresponding l slices are reassembled into a new sequence, repeating this random computation process N times. In this study, we set the l to 128, with each slice containing 32 values, and N was set to 20/50. Once the model training was completed, two models were randomly coded with noise 20/50 times, respectively, to generate test data.

Finally, the extended results obtained from the two traditional methods and the WPTSE-Net undergo a reverse normalization process to yield the ultimate simulated outcomes. Subsequently, all these outcomes are meticulously analyzed in comparison with the ground truth, focusing on statistical characteristics, spectral properties, peak values, etc.

2.4. Evaluation metrics for time series extension

Unlike predictive or fitting methods, generative methods do not directly evaluate the similarity between generated sequences and true values. To objectively assess the outcomes of these generative

methods, this study employs assessments from three perspectives: statistical characteristics, spectrum analysis, and peak value distribution.

Several statistical metrics were employed, including mean, variance, skewness, and kurtosis. The mean is calculated as the sum of all values divided by the total number of values. Variance measures the dispersion of data points around the mean. It is calculated by taking the average squared differences between each data point and the mean. Skewness measures the asymmetry of the distribution. A positive skewness indicates a longer tail on the right side, while a negative skewness indicates a longer tail on the left side. Kurtosis quantifies the shape of the distribution and measures the presence of outliers or extreme values. It is calculated by comparing the tails of the distribution to that of a normal distribution. The specific calculation methods for these metrics are shown as Eq. (11) to Eq. (14):

$$\mu(y) = \frac{\sum_{i=1}^n y}{n} \quad (11)$$

$$\sigma(y) = \sqrt{\frac{\sum_{i=1}^n (y - \mu(y))^2}{n - 1}} \quad (12)$$

$$\gamma_3(y) = \frac{\frac{1}{n} \sum_{i=1}^n (y - \mu(y))^3}{\sigma(y)^3} \quad (13)$$

$$\gamma_4(y) = \frac{\frac{1}{n} \sum_{i=1}^n (y - \mu(y))^4}{\sigma(y)^4} \quad (14)$$

where y represents the data to be calculated, and n represents the length of data.

The power spectrum primarily captures the relationship between different time steps at the same spatial point, while being relatively insensitive to spatial variations on the building's surface, making it suitable for examining differences between various methods. It is often computed using a Fourier transform or other spectrum analysis techniques. The calculation formula for the power spectrum is indicated in Eq. (15). In this study, we particularly emphasize the comparison of bispectra. The bispectrum is a powerful tool for analyzing stationary random signals that are nonlinear, non-Gaussian, and non-minimum phase, and can be used to reveal features such as resonances, nonlinear distortions, and other features in the signal. The calculation formula for the bispectrum is indicated in Eq. (16).

$$\phi_{yy}(w) = FFT(\phi_{yy}(\tau)) \quad (15)$$

$$\phi_{yy}(w_1, w_2) = FFT(\phi_{yy}(\tau_1, \tau_2)) \quad (16)$$

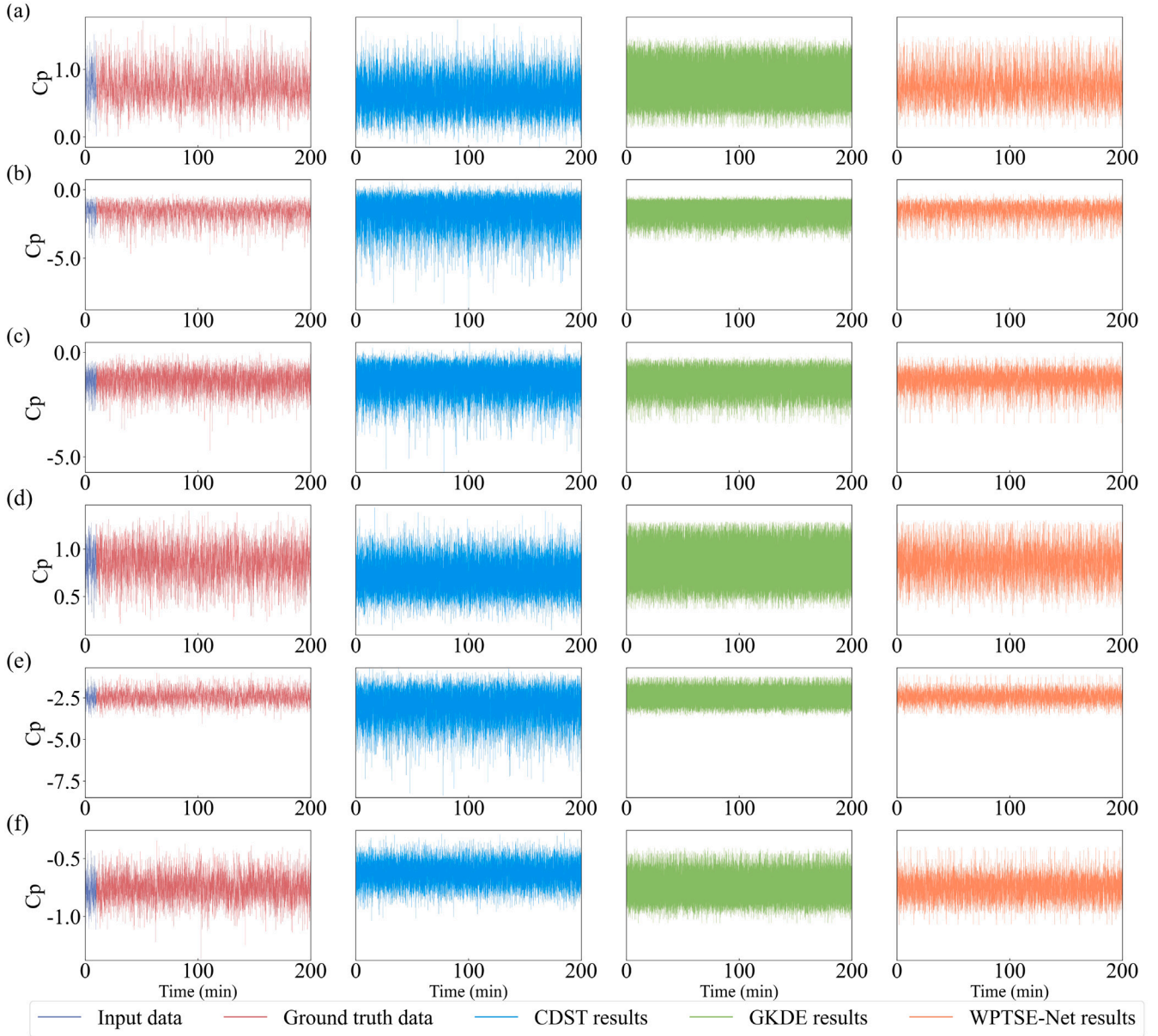


Fig. 5. Visualization of ground truth and extended data (C_p): (a) Point 1; (b) Point 2; (c) Point 3; (d) Point 4; (e) Point 5; (f) Point 6.

$$\phi_{yy}(\tau_1, \tau_2) = E(y(t)y(t + \tau_1)y(t + \tau_2)) \quad (17)$$

where $\phi_{yy}(\tau)$ represents the autocorrelation function of data y , and FFT represents the Fourier Transform operation. In Eq. (16), τ represents the higher-order autocorrelation function of y , and the specific calculation formula is given by Eq. (17).

Peak values encompass the maximum, minimum, and extreme values observed in a dataset. They provide valuable insights into the range of values present in the dataset and offer a deeper understanding of potential exceptional or anomalous observations, which can be crucial for data analysis and decision-making processes. Consequently, this study also compares the local maximum and minimum values obtained from extended results. Furthermore, two commonly used extreme value distributions in engineering, the Gumbel distribution, and the Generalized Extreme Value (GEV) distribution are introduced for evaluating and contrasting extreme values, as expressed in Eqs. (18)–(19):

$$f(x_t; \Phi, \beta) = \frac{1}{\beta} \exp \left[-\frac{x_t - \Phi}{\beta} - \exp \left(-\frac{x_t - \Phi}{\beta} \right) \right] \quad (18)$$

$$f(x_t; \Phi, \sigma, \xi) = \frac{1}{\sigma} \left[1 + \xi \left(\frac{x_t - \Phi}{\sigma} \right) \right]^{-1/\xi-1} \exp \left[- \left(1 + \xi \left(\frac{x_t - \Phi}{\sigma} \right) \right)^{-1/\xi} \right] \quad (19)$$

where Φ is the location parameter, β is the scale parameter, and ξ is the shape parameter.

3. Results and discussions

This study assumes that the full-length data obtained from the wind tunnel tests represents the ground truth. The statistical characteristics, spectra, and extremes derived from these data are considered stable and serve as a reliable benchmark for comparison with the extended results simulated by various methods.

3.1. Visualization of extended time series data

Fig. 5 shows the extended time series data of C_p obtained through the three methods: CDST, GKDE, and WPTSE-Net. It can be seen that

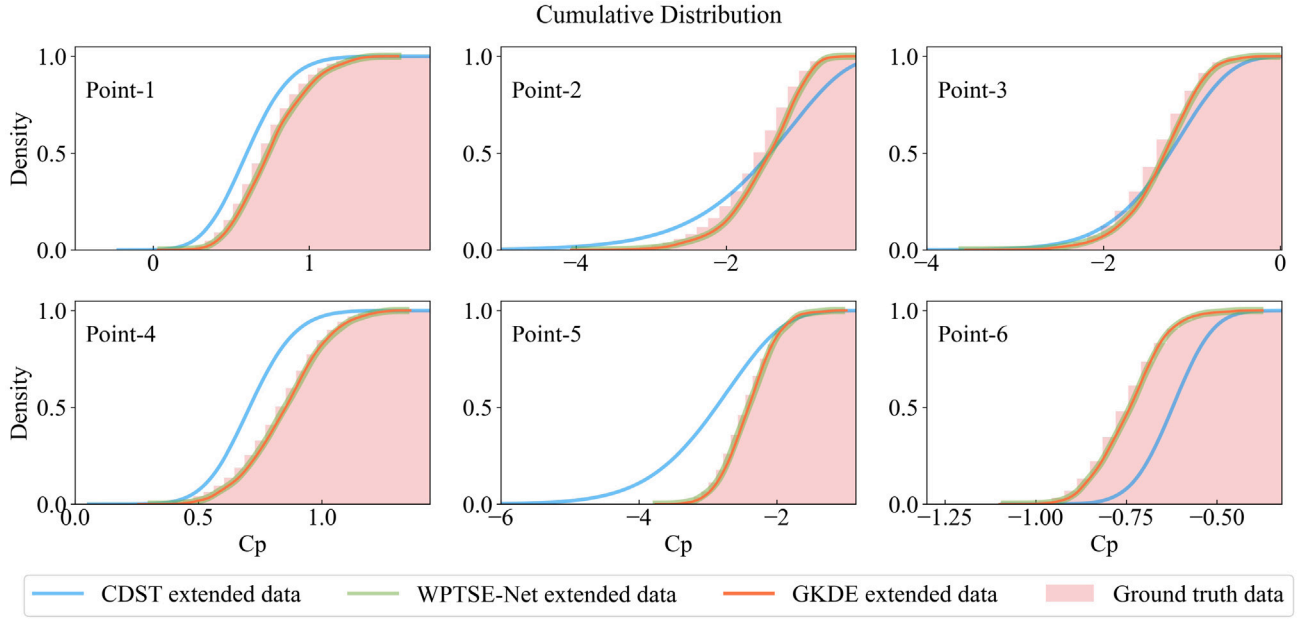


Fig. 6. Cumulative distribution of ground truth and extended data.

the CDST method approximates the true values at the front measurement points (points 1 and 4). However, it shows poor performance at the side measurement points (points 3 and 6) and fails to accurately represent the characteristics of the true values at the corner measurement points (points 2 and 5). For the GKDE method, the extended results are consistent with the numerical range of the ground truth at all points. However, the GKDE results show excessive stability (low variance) to the input data, indicating significant overfitting. In direct comparison with the two previous methods, the WPTSE-Net results are superior in terms of numerical range, variation, and trends.

In practical scenarios, wind pressure time series data exhibit distinct distribution properties across different surfaces of buildings. The windward surface typically displays positive pressure, characterized by a relatively simple distribution of C_p , often resembling a Gaussian distribution. In contrast, the distribution of C_p on the lateral surface is more complex, exhibiting phenomena such as vortices and separation. This complexity makes it difficult to describe the pressure distribution in these regions using a Gaussian distribution. For buildings with chamfered edges, extreme wind pressures can randomly occur in the separated flow regions along the leading edge, the leeward side, and the chamfered areas on the windward side. This leads to pronounced non-Gaussian characteristics in the wind pressure distribution. Factors such as edge flow on the windward side, flow separation on the leeward side, and reattachment effects contribute to these non-Gaussian pulsations. These airflow patterns at the corners of bluff bodies are further complicated. From an intuitive comparison of the extended results, the performance of the CDST method is significantly affected by the distribution properties of C_p . In contrast, the results of the GKDE method are overly conservative and less responsive to distribution characteristics. The WPTSE-Net methods, however, excel in handling both Gaussian and non-Gaussian distribution data, making them the preferable choice for simulating C_p in the whole building.

The cumulative distribution of the three methods provides further confirmation that WPTSE-Net is the most effective and stable approach. Fig. 6 indicates that the cumulative distribution of GKDE and WPTSE-Net closely corresponds to the ground truth data. Conversely, the outcomes of CDST exhibit substantial deviations from the true values, particularly in the extended data for corner regions (points 2 and 5). In terms of cumulative distribution curves, an analysis of the extension outcomes between Models I and II indicates that the CDST method performance is significantly influenced by alterations in

the shape of the corners. In contrast, the other two methods exhibit minimal responsiveness to such modifications. Both GKDE and WPTSE-Net exhibit nearly identical trends and slopes that align with the ground truth. Overall, WPTSE-Net demonstrates an advantage in ultra-time simulation.

In addition, we performed a thorough analysis of a particular subset, specifically the first ten-minutes of the data. Fig. 7 illustrates the comparison between the three methods and the ground truth. The results obtained through the CDST method demonstrate notable discrepancies, particularly in the numerical range and mean values. Meanwhile, the time series extension generated by GKDE displayed excessively dense fluctuations that did not align with the expected characteristics. This observation suggests that the GKDE method treats C_p data as isolated points rather than a continuous time series, thereby disregarding the intrinsic interconnectivity between these points. In contrast to the other two methods, the extended results from WPTSE-Net demonstrate superior similarity and continuity to the ground truth. This suggests that WPTSE-Net has good simulation ability and effectively preserves the temporal relationships within the wind pressure time series data, emphasizing a more accurate representation of the continuous nature of C_p .

Following a comprehensive examination of the overall and localized results, it can be asserted that WPTSE-Net exhibits superior modeling capabilities in comparison to CDST and GKDE. WPTSE-Net is capable of handling both non-Gaussian and Gaussian distribution data, consistently producing simulation results that are in close alignment with the ground truth. Importantly, WPTSE-Net maintains strong generalization without compromising the continuity between data points.

3.2. Statistical characteristics

To quantitatively evaluate the three methods, we compared their results in the first four order moments: mean, variance, skewness, and kurtosis. These characteristics are shown in Fig. 8. The assessment of mean values indicates that both GKDE and WPTSE-Net are in close alignment with the true values at all points, whereas the CDST method consistently yields lower values than the true values. With regard to the variance, both GKDE and WPTSE-Net demonstrate a high degree of concordance with the ground truth. However, CDST exhibits notable discrepancies, particularly at two corner points (points 2 and 5). The higher-order indicators reveal significant differences, especially in

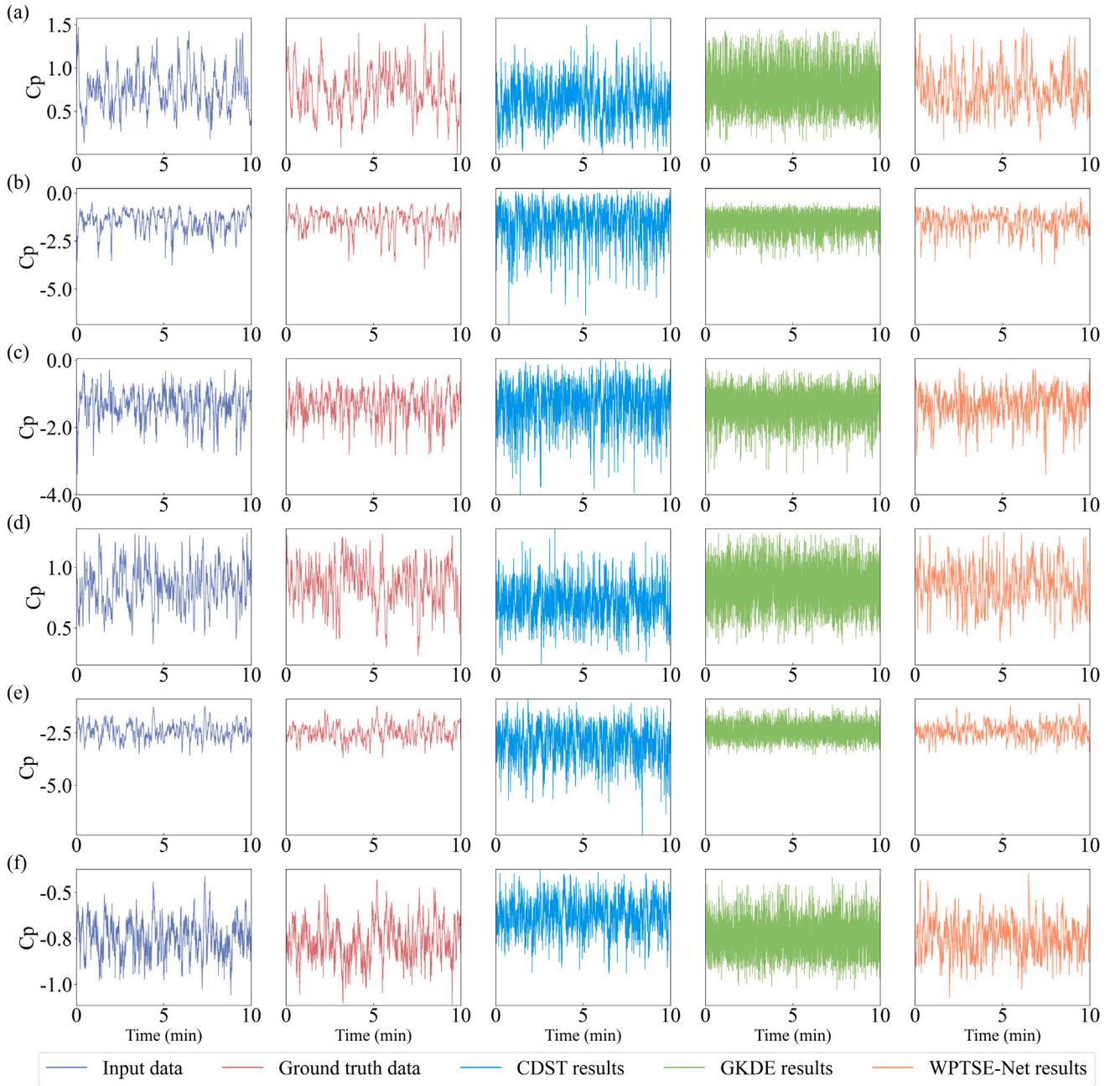


Fig. 7. Visualization of first ten-minute subset via different methods: (a) Point 1; (b) Point 2; (c) Point 3; (d) Point 4; (e) Point 5; (f) Point 6.

Model II, where CDST shows indicators that are almost the opposite of the true values at each point. The extended results of GKDE and WPTSE-Net demonstrate comparable advantages in skewness and kurtosis, with WPTSE-Net slightly outperforming Model II.

CDST performs well in capturing mean values but shows weaker results in other indicators, particularly in higher-order characteristics at corner points. This highlights the significant impact of building shape on CDST performance and its challenge in capturing non-Gaussian sequence features. Consequently, CDST may face limitations when applied to structurally complex building shapes. In contrast, WPTSE-Net and GKDE show results that are closer to the ground truth in the first- and second-order moment characteristics. Notably, WPTSE-Net exhibits superior performance in higher-order indicators. For example, WPTSE-Net consistently demonstrates stable performance and excels in higher-order indicators in Model II, confirming its superior stability compared to GKDE.

The analysis presented in this section serves to reinforce the conclusions drawn in Section 3.1, confirming that GKDE outperforms CDST, while WPTSE-Net is superior to GKDE. In summary, both from an intuitive assessment of the data and through a quantitative analysis of its characteristics and patterns, WPTSE-Net consistently demonstrates superior overall performance compared to the other two methods.

3.3. Spectrum analysis

The power spectrum and bispectrum are important metrics for evaluating wind pressure time series simulations. They provide insights into data characteristics from a frequency-domain perspective and facilitate the analysis of results across different frequency components. Fig. 9 presents the power spectrum density plots for six measurement points. The power spectrum generated by the CDST method is suboptimal, as it underestimates power in the low-frequency range while overestimating

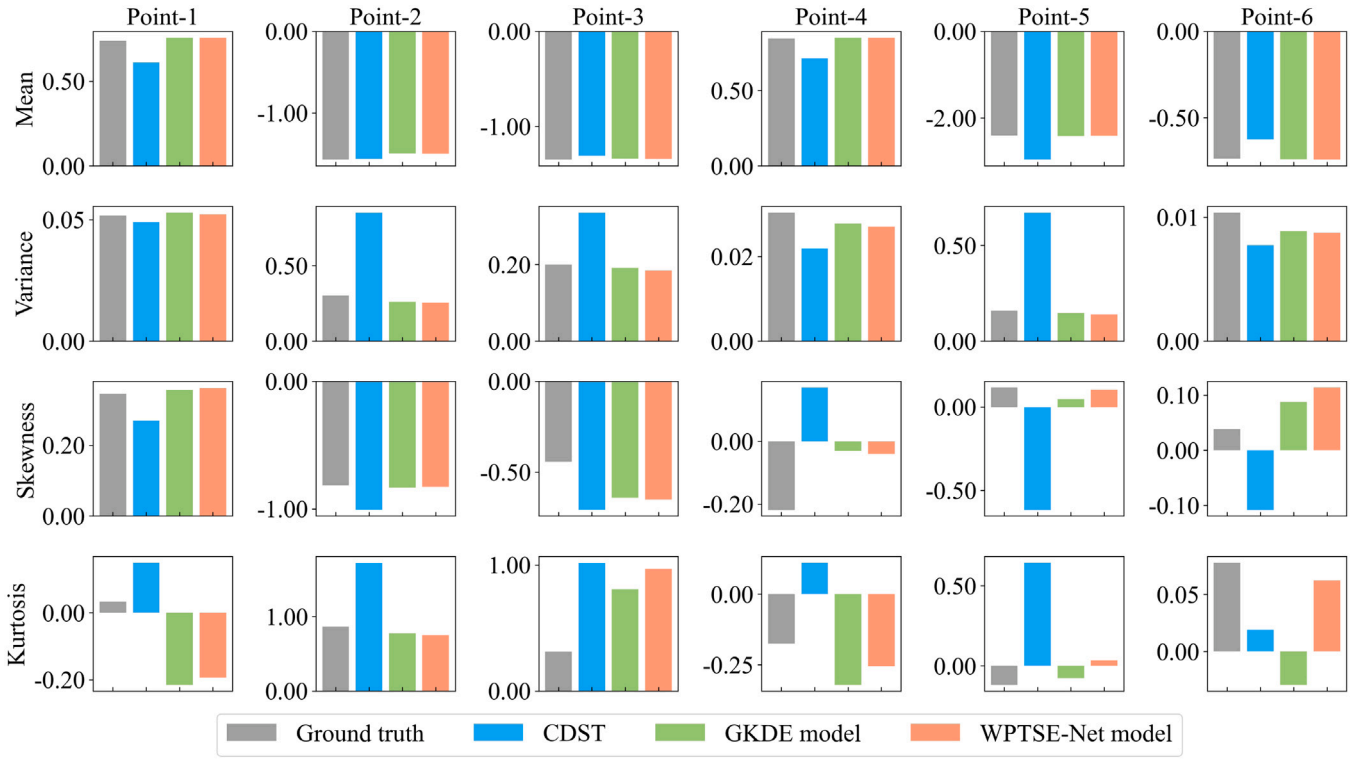


Fig. 8. Statistical properties of ground truth and extended data.

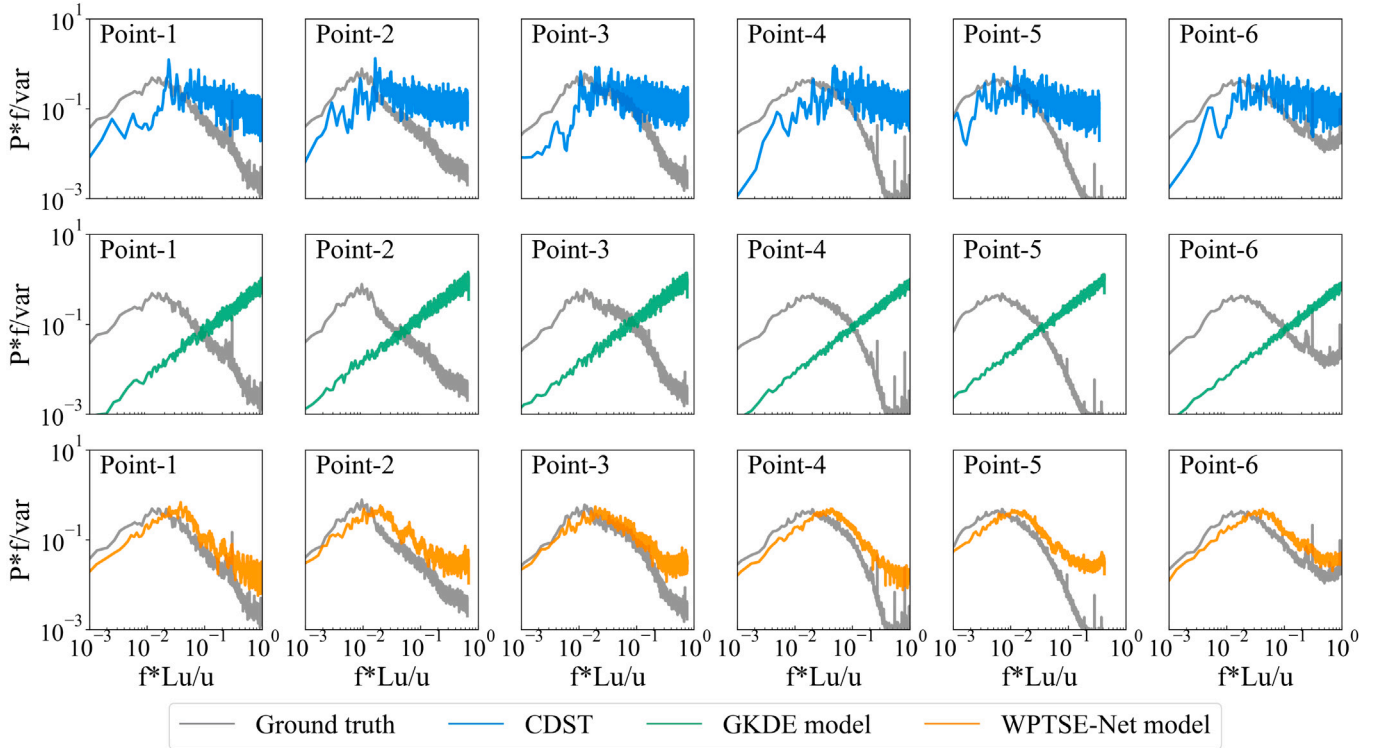


Fig. 9. Power spectrum density of true values and extended results.

it in the high-frequency range. The extended results from GKDE display a linear pattern in the power spectrum density curve, which deviates significantly from the ground truth. This discrepancy suggests that GKDE's results exhibit different signal characteristics or energy distri-

butions compared to the actual data, indicating that GKDE may not accurately capture the C_p values of the two building models. Although WPTSE-Net slightly overestimates power in the high-frequency region, its peak is closer to the true value, and its overall trend and pattern

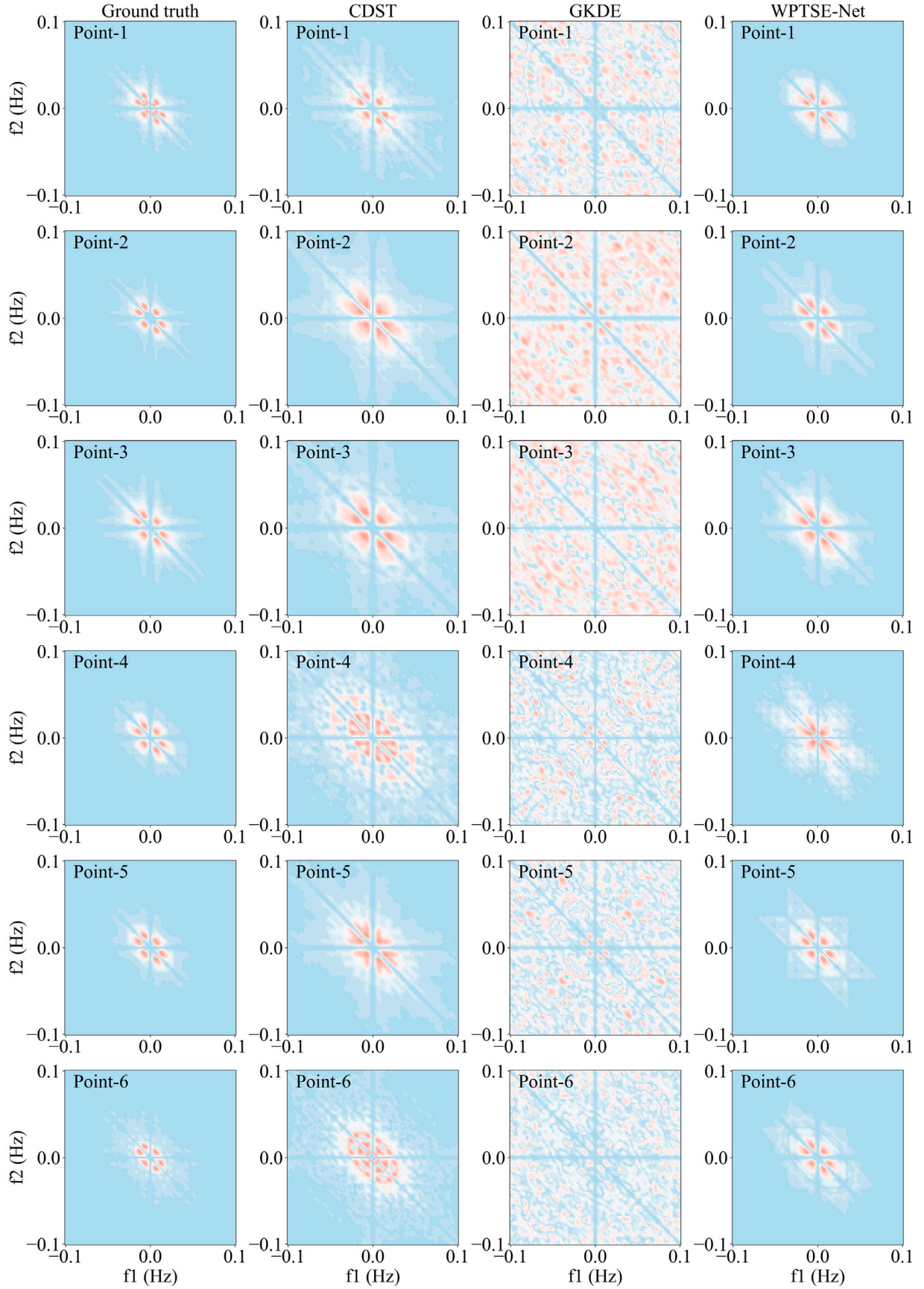


Fig. 10. Bispectrum analysis of ground truth values and extended data.

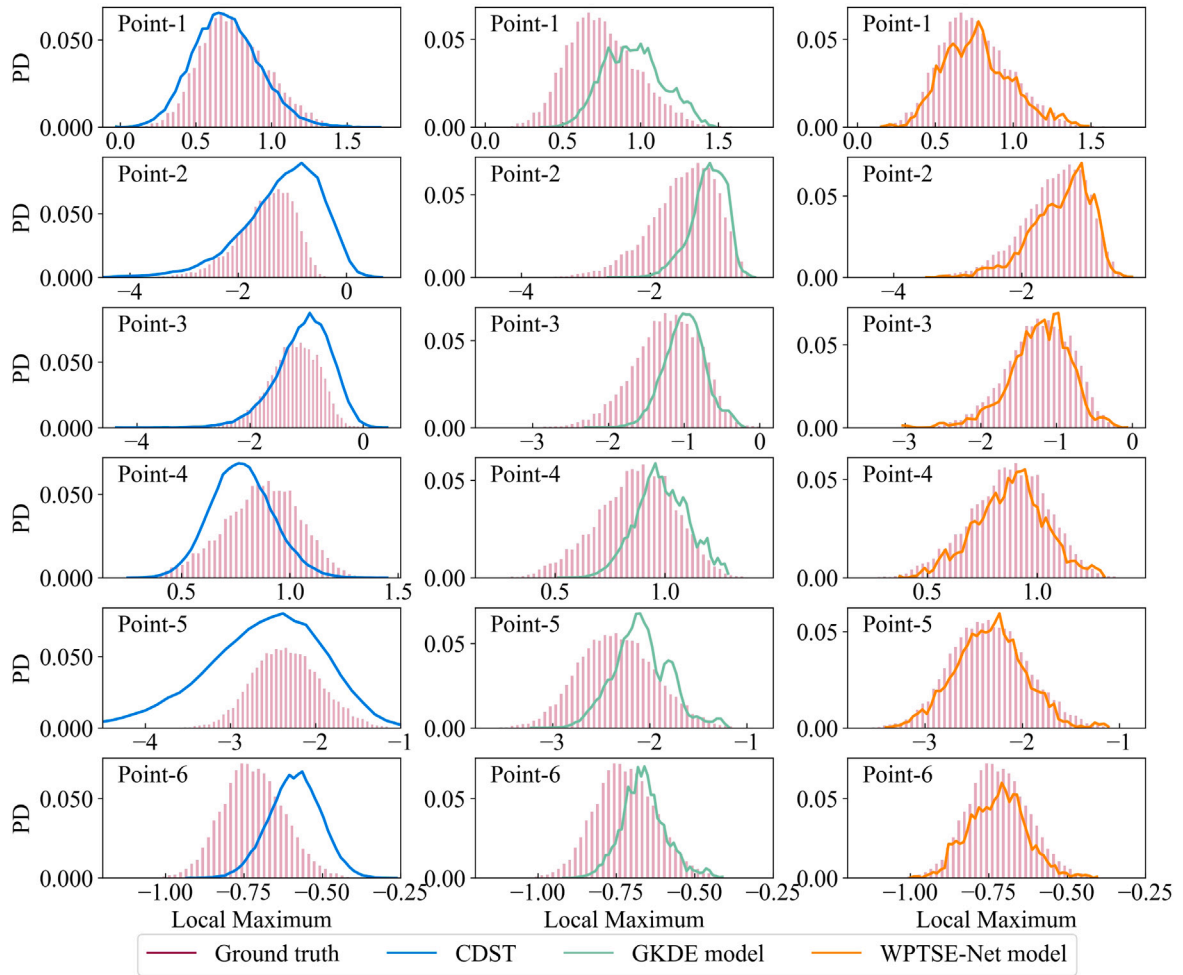


Fig. 11. Probability density distributions of local maximum in extended results obtained through different methods.

align more closely with the actual power spectrum density than CDST. This demonstrates WPTSE-Net's superior ability to capture the spectral characteristics of the data, outperforming both CDST and GKDE.

Compared to the power spectrum, the bispectrum focuses more on analyzing non-linear relationships and higher-order characteristics. In Fig. 10, a comprehensive analysis using the bispectrum metric reveals distinct performance characteristics between different methods. Specifically, the bispectral maps generated by WPTSE-Net, depicting nonlinear interaction strengths, concentrate in similar regions. This observation suggests that corresponding frequency combinations exhibit similar non-Gaussian or non-linear behavior in their signals, emphasizing the effectiveness of WPTSE-Net in capturing the complex frequency structure and signal patterns of the ground truth. CDST shows too many diffuse non-linear interactions between different frequency components at different measurement points, with local deviations from the ground truth, despite some similarity in certain frequency pairs. In contrast, CDST is relatively weak in capturing similar nonlinear relationships in the frequency domain, and it is slightly less stable. Regarding the GKDE method, its extension's results perform significantly poorly in the bispectrum and suffer from severe distortions. This suggests that GKDE may have inherent limitations, particularly in representing the true nonlinear interactions between frequency pairs in the wind pressure time series. As previously inferred, due to the approach treating wind pressure values independently at each time step during the extension process, leads to irregular reflections of the strength of nonlinear interactions under different frequency combinations. As Gurley et al. (1996) observed, most static transfer action-based schemes fail to capture the contribution of embedded memory, whereas ANN-based simulations,

like WPTSE-Net, successfully represent memory features, as evidenced in the bispectrum comparison.

In the spectral analysis comparison, WPTSE-Net emerges as the most effective method for both power spectrum and bispectrum analysis, outperforming CDST and GKDE. Thus, in terms of the statistical and frequency-domain distribution of the simulation results, WPTSE-Net proves to be the most realistic choice among the three methods.

3.4. Peak values assessment

Finally, we conducted an in-depth analysis of the peak value distribution in the extension results. Figs. 11 and 12 illustrate the distribution curves for local minima and local maxima, respectively. The results depicted in the figures reveal that the extended data from WPTSE-Net closely aligns with the ground truth data within the intermediate range of C_p . However, CDST often tends to overestimate the distribution of different peak values, which appears to be unreasonable. As for the GKDE results, they exhibit a fitting performance far inferior to WPTSE-Net, with noticeable deviations in the simulated local maximum and local minimum.

Extreme values are crucial to ensure the safe operation of structures under the most unfavorable conditions and must be carefully examined. In this section, extreme values were extracted from the results of three methods, and a comparative analysis was performed by estimating them using Gumbel and GEV distributions. The calculated extreme values for ten-minute return periods are presented in Table 1. CDST estimates prioritize safety by adopting a cautious approach. However, they tend to be overly conservative, particularly for the corners of bluff

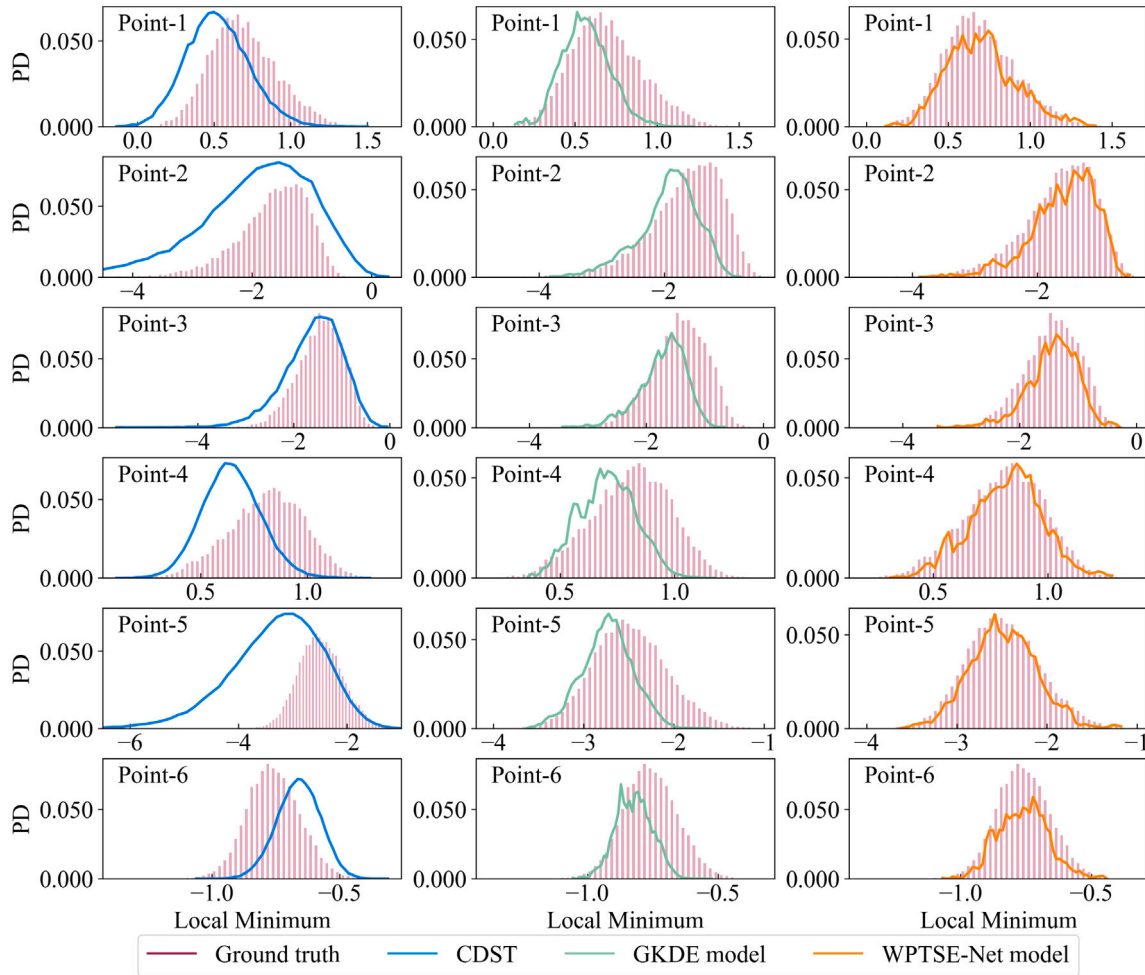


Fig. 12. Probability density distributions of local minimum in extended results obtained through different methods.

Table 1

Extremes values of ground truth and extended results via through different methods.

Point number	X_{True}	X_{CDST}	X_{GKDE}	X_{WPTSE}	X_{Gumbel}	X_{GEV}
Point-1	1.773	1.736	1.496	1.499	1.530	1.640
Point-2	4.814	8.790	3.858	3.908	3.173	2.780
Point-3	4.704	5.751	3.463	3.430	2.855	2.719
Point-4	1.438	1.463	1.297	1.315	1.405	1.322
Point-5	4.058	8.500	3.648	3.656	3.691	3.702
Point-6	1.379	1.070	1.061	1.075	1.081	1.101

bodies (Points 2 and 5), and lack robustness. Both GKDE and WPTSE-Net produce lower estimations, which is an inherent trait of data-driven algorithms. In data-driven algorithms, a high percentage of samples tends to dominate the direction of error optimization. Furthermore, the DL model has a larger number of weight update gradient components for high percentage samples that can potentially exacerbate such inequalities (Anand et al., 1993; Guo et al., 2017; Johnson and Khoshgoftaar, 2019). Therefore, GKDE and WPTSE-Net models excel in analyzing and recognizing frequently occurring data, such as values within the intermediate range. Conversely, extreme values are less common in the entire dataset, which limits their exposure during the model training process. Nevertheless, from the perspective of numerical errors alone, the WPTSE-Net remains the best among the three methods and also outperforms other traditional extreme value assessment methods.

In light of the comparison, it is crucial to acknowledge that the extended extreme values should be treated as references rather than precise values. Practically, we suggest applying a designated safety

factor to WPTSE-Net results, aiming for a balance between safety considerations and economic efficiency in decision-making.

4. Conclusions

In this study, we introduced a DL model Wind Pressure Time Series Extend Network (WPTSE-Net) for extending C_p time series data. Unlike classical generative models, WPTSE-Net reconstructs the encoder and incorporates prior knowledge, significantly reducing data requirements while ensuring robust generalization capabilities. We conducted an in-depth analysis of the WPTSE-Net model performance across three aspects, statistical metrics, spectral description, and extreme values.

Results indicate WPTSE-Net's superiority across most metrics compared to traditional methods like Correlation Distortion Static Transform (CDST) and Gaussian Kernel Density Estimation (GKDE). Specifically, WPTSE-Net closely aligns with the ground truth data in capturing time series fluctuations and continuity, excels in determining mean and variance accurately, and outperforms other methods in skewness and kurtosis determination and maintaining the embedded memory. In spectral analysis, WPTSE-Net remains optimal, displaying better consistency with actual power spectra, frequency structure, and pattern similarity. Furthermore, the quality of fitting of WPTSE-Net simulation data is the highest when examining the peak values distribution of C_p values. Additionally, the extreme values generated by WPTSE-Net are numerically the closest to the true values among the methods evaluated. These findings underscore WPTSE-Net's effectiveness in enhancing the accuracy and reliability of C_p time series extension.

Although this study has demonstrated the superiority of WPTSE-Net models over traditional methods in extending the wind pressure time series, several aspects warrant further investigation. Firstly, despite the nominal accuracy improvement of the current WPTSE-Net model, its extended results should be regarded as reference values in engineering practice. To balance efficiency and safety, it is advisable to leverage the extended results of WPTSE-Net and scale them by a multiplication factor. Secondly, there is room for improvement by simultaneously considering correlated processes that is the interactions of multiple measurement points and introducing correlation variables between them. The study focused solely on data from measurement points at the same altitude; however, in reality, patterns and connections exist between points in different planes at various altitudes. Incorporating multiple measurement points could enhance the model's performance and robustness. Thirdly, the theoretical enhancement of WPTSE-Net models could be achieved through leveraging transfer learning to improve the model's performance on Cp data. For instance, utilization of knowledge from analogous but different domains such as aerodynamics, fluid dynamics, structural dynamics, and vibration-related data, as well as engineering feature extraction techniques from the field of signal processing, etc.

CRedit authorship contribution statement

Biao Tong: Writing – original draft, Visualization, Methodology, Investigation, Formal analysis. **Yang Liang:** Writing – review & editing, Visualization, Methodology, Investigation. **Jie Song:** Supervision, Methodology, Data curation. **Gang Hu:** Writing – review & editing, Supervision, Project administration, Funding acquisition, Formal analysis. **Ahsan Kareem:** Writing – review & editing, Supervision, Methodology, Conceptualization.

Declaration of competing interest

We declare that no conflict of interest exists in the submission of this manuscript. All authors have approved the manuscript for publication. We affirm that the research described in this manuscript is original and has not been published previously, nor is it under consideration for publication elsewhere, either in whole or in part. The enclosed manuscript has been reviewed and approved by all authors.

Data availability

The authors do not have permission to share data.

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